Mechanisms & Vibrations

- Introduction to Mechanisms

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Course outline

- Mechanism overview
- Terminologies
- Kinematic diagrams
- Linkages

Mechanism assignment #1: Kinematic diagrams

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Mechanism assignment #1: Kinematic diagrams

Mechanism

- Solid mechanics has three major subbranches:
  - Kinematics: the study of motion, without regard to forces.
  - Statics: the study of forces and moments, apart from motion
  - Kinetics: the study of forces on systems in motion
- The combination of kinematics and kinetics is dynamics.
- We discuss kinematics and dynamics required to accomplish mechanism design.

Mechanism Overview

- Mechanism: a mechanical device that has the purpose of transferring motion and/or force from a source to an output.

- A mechanical device: A moveable structure made of two basic types of components:
  - Mechanical parts (solid body)
  - Joints (joints/actuators)

Mechanism Overview (Cont.)

- Linkage
  - Consists of links (or bars)
  - Links: rigid or flexible members with at least two nodes (points of attachment).
  - Considered rigid (generally)
  - Connected by joints (e.g. pins, revolute, prismatic)
  - Form open or closed chains (or loops)
- Kinemtaic chains, with at least one link fixed, become frame
  - Mechanism: If at least two other links retain mobility → input & output
  - Structure: If there is no mobility

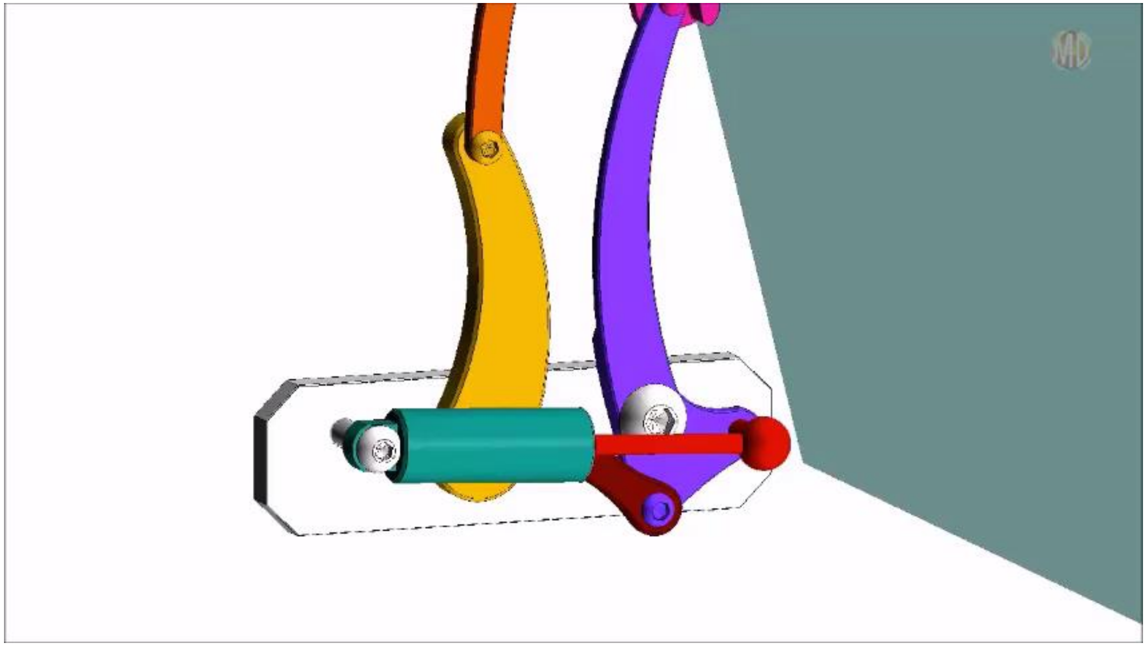
Mechanism Examples

- Car hood

Source: classicindustries.com

Mechanism Examples

- Car hood



Source: https://youtu.be/jc057w0qTNQ

Mechanism Examples (Cont.)

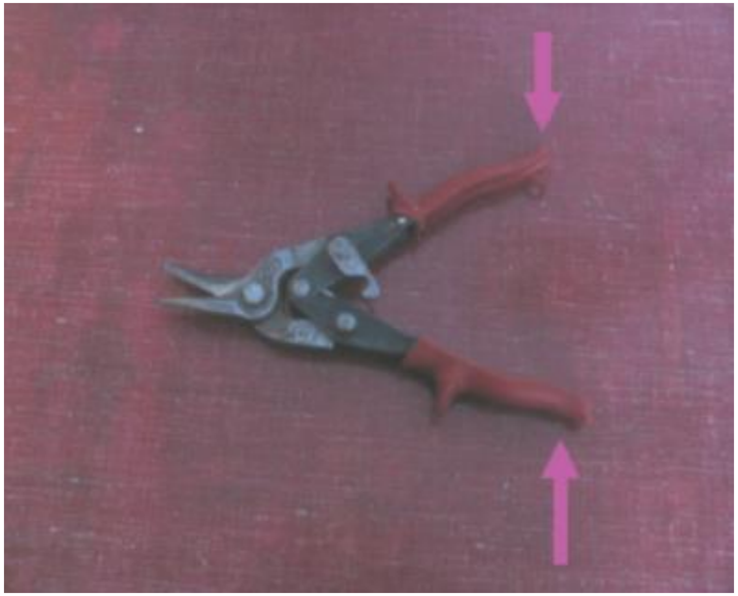
- Folding chair for motion transfer

Open

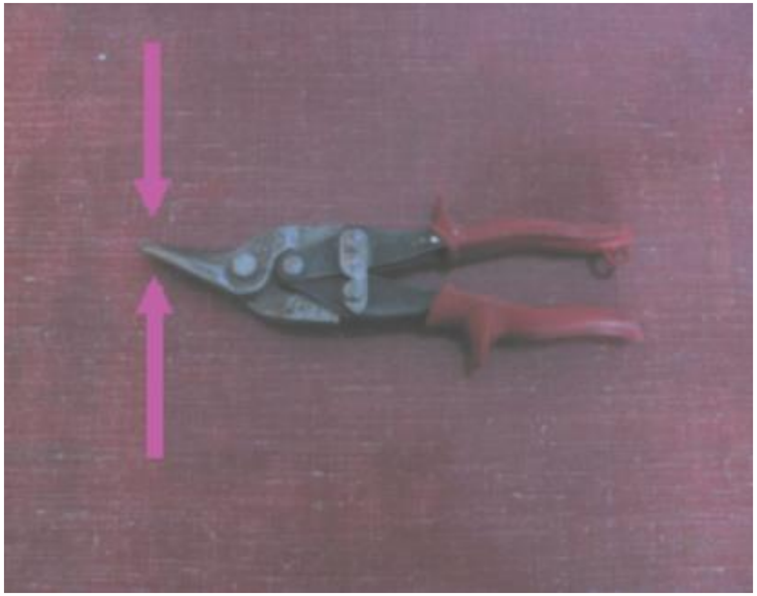
Closed

Mechanism Examples (Cont.)

- Scissors for force transfer



Input force



Output force

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Mechanism assignment #1: Kinematic diagrams

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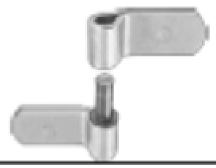
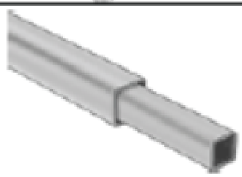
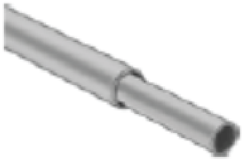
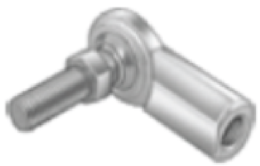
Terminologies

- Link: a generally rigid body linked by a joint.
- Joint: in general called kinematic pair, providing connection between links that permits constrained relative motion.
- Lower pair: surface contact.
- Higher pair: line and point contact.
- Planar: move in 2-dimensional plane.
- Spatial: move in 3-dimensional plane.

Terminologies (cont.)

- Linkages: referred to kinematic chains joined by only lower pairs.
- Higher pair: line and point contact.
- Mechanisms: spatial linkages including both lower and higher pairs.
- Open chain: one or more of the links is connected to only one other link.
- Closed chain: each link is connected to two or more other links
- The degrees of freedom (dof): the number of independent coordinates required to uniquely define its position.

Lower pair joints

Joint	Symbol	DOF	Note
	R	1	Elementary
	P	1	Elementary
	U	2	R+R, ⊥
	C	2	R+P, //
	S	3	R+R+R, ⊥

https://www.brighthubengineering.com/machine-design/6577-types-of-kinematic-joints/

Open chain example

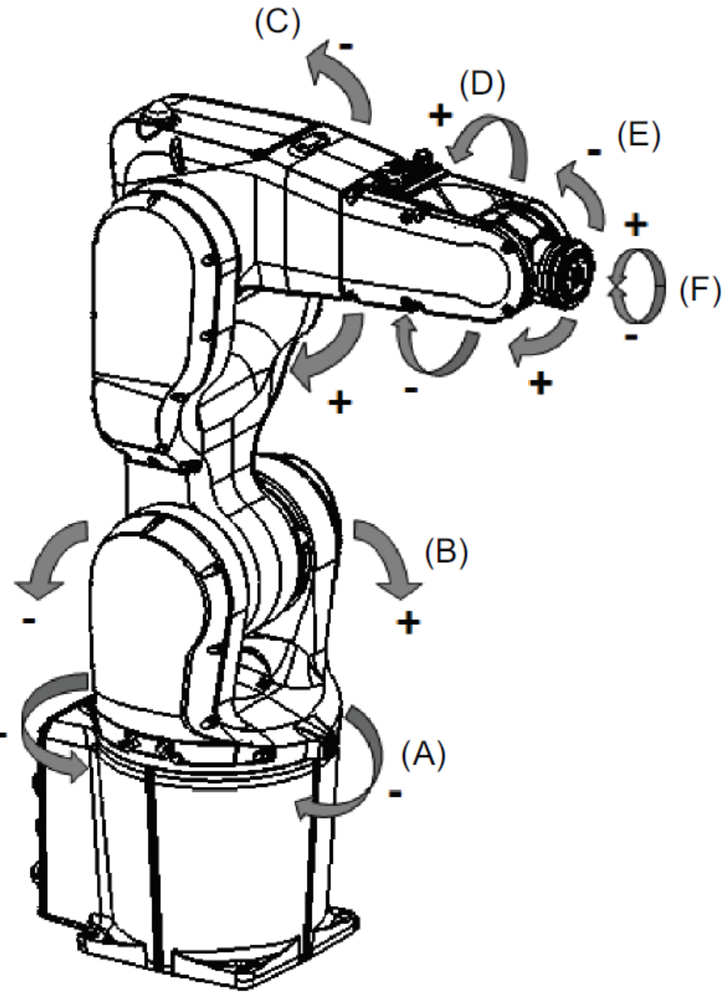
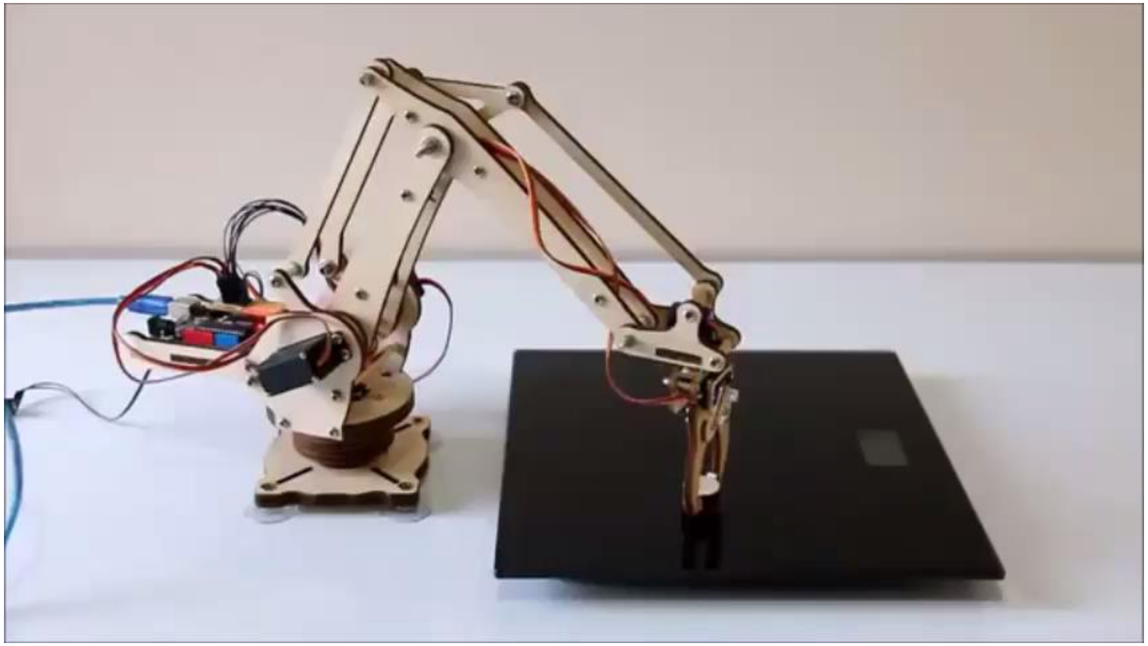


ABB IRB 1200 industrial robot

Open chain example



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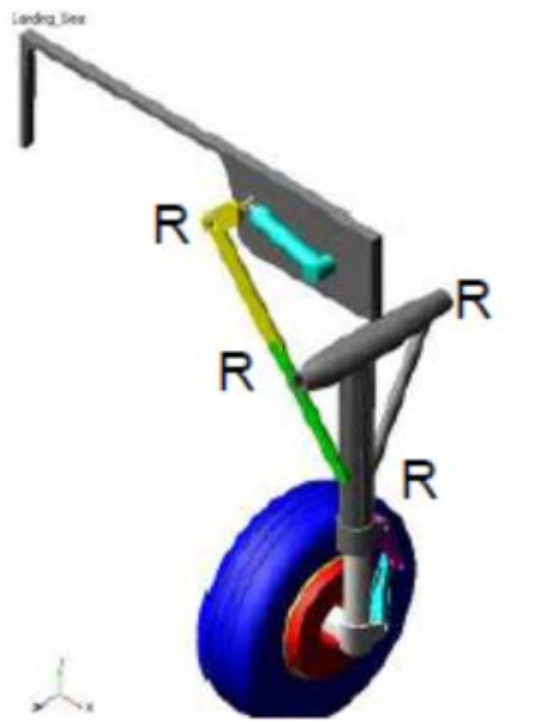
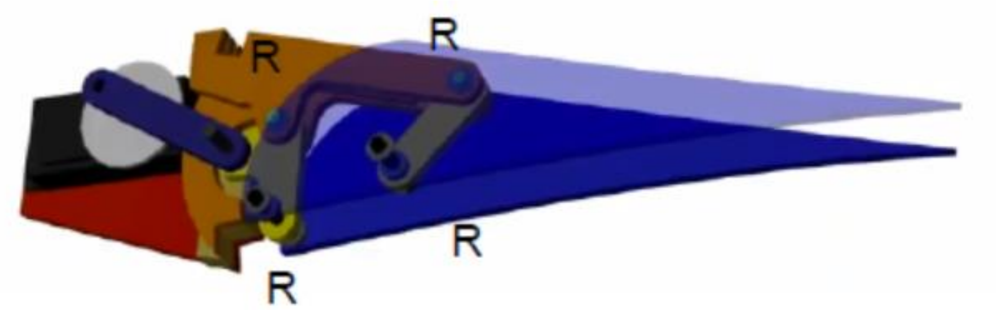
Closed chain exmple

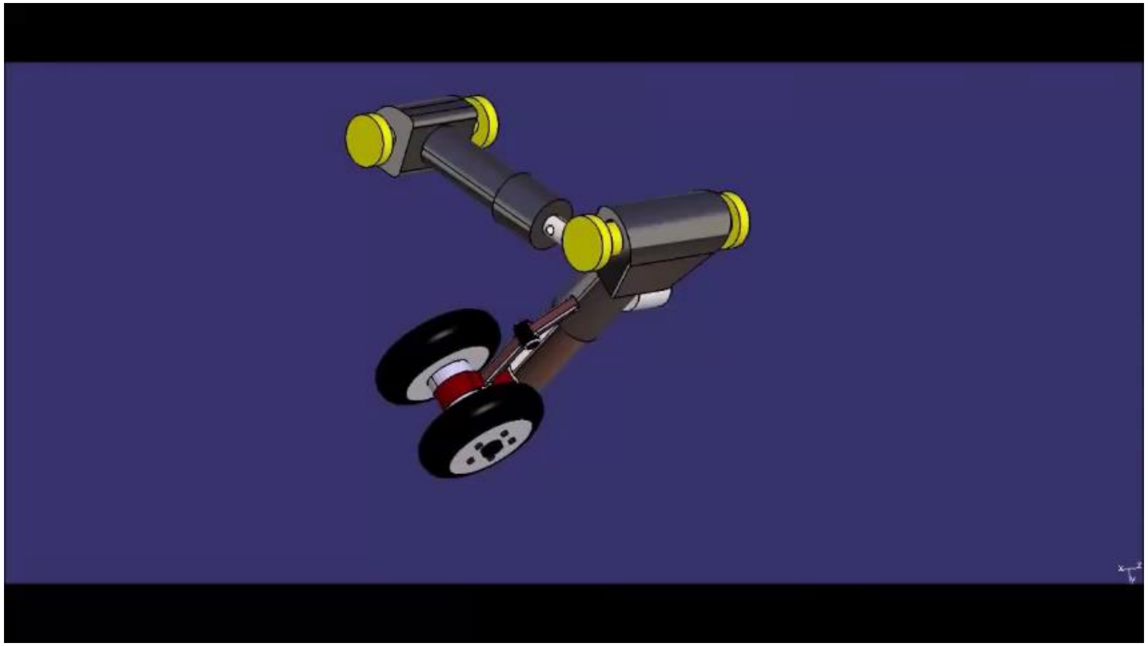


Landing gear development



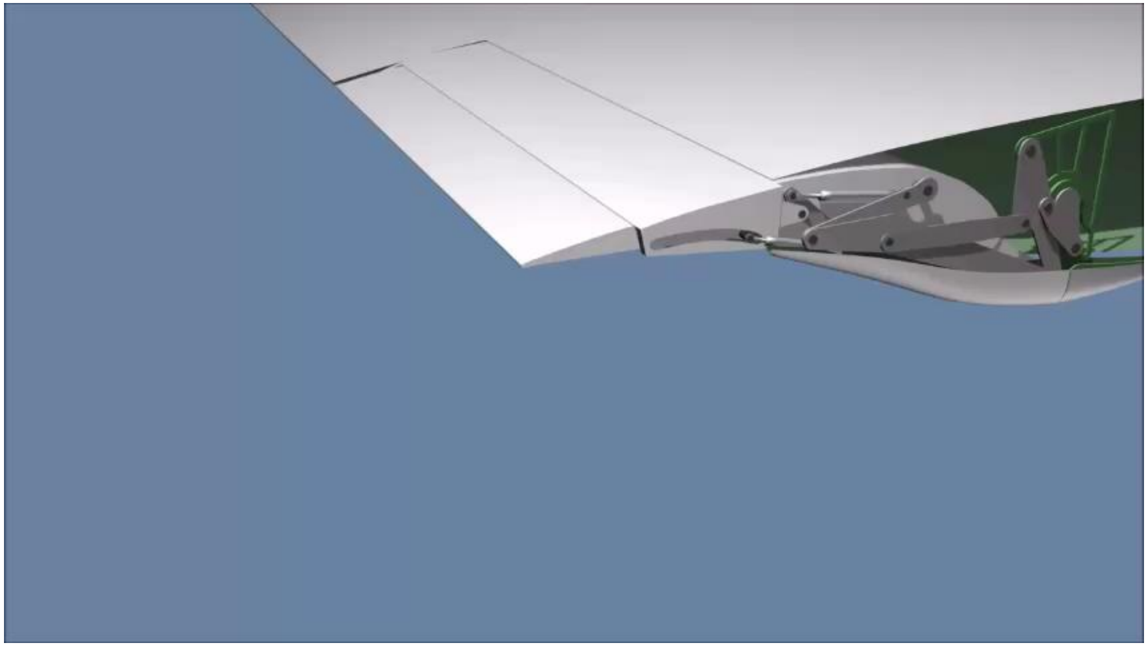
Flap development

Closed chain example



Source: https://youtu.be/st-2A1T6vrI

Closed chain example



Source: https://youtu.be/kKDMjc3l_gw

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Mechanism assignment #1: Kinematic diagrams

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Kinematic diagrams

- Also called skeleton or stick diagram.
- Represent the essential structure (skeleton) of a mechanism, similar idea to electrical circuit diagram.
- (Basic symbol) lines or simple shapes are used to represent links labelled by numbers.
- (Basic symbol) circles or blocks are used to represent joints labelled by letters (capital).

Kinematic diagrams (Cont.)

- Can be scaled or non-scaled (distance between two joints).
- Non-scaled diagrams are used for mobility (DOF) analysis.
- Scaled diagrams can be used for motion and force analysis.

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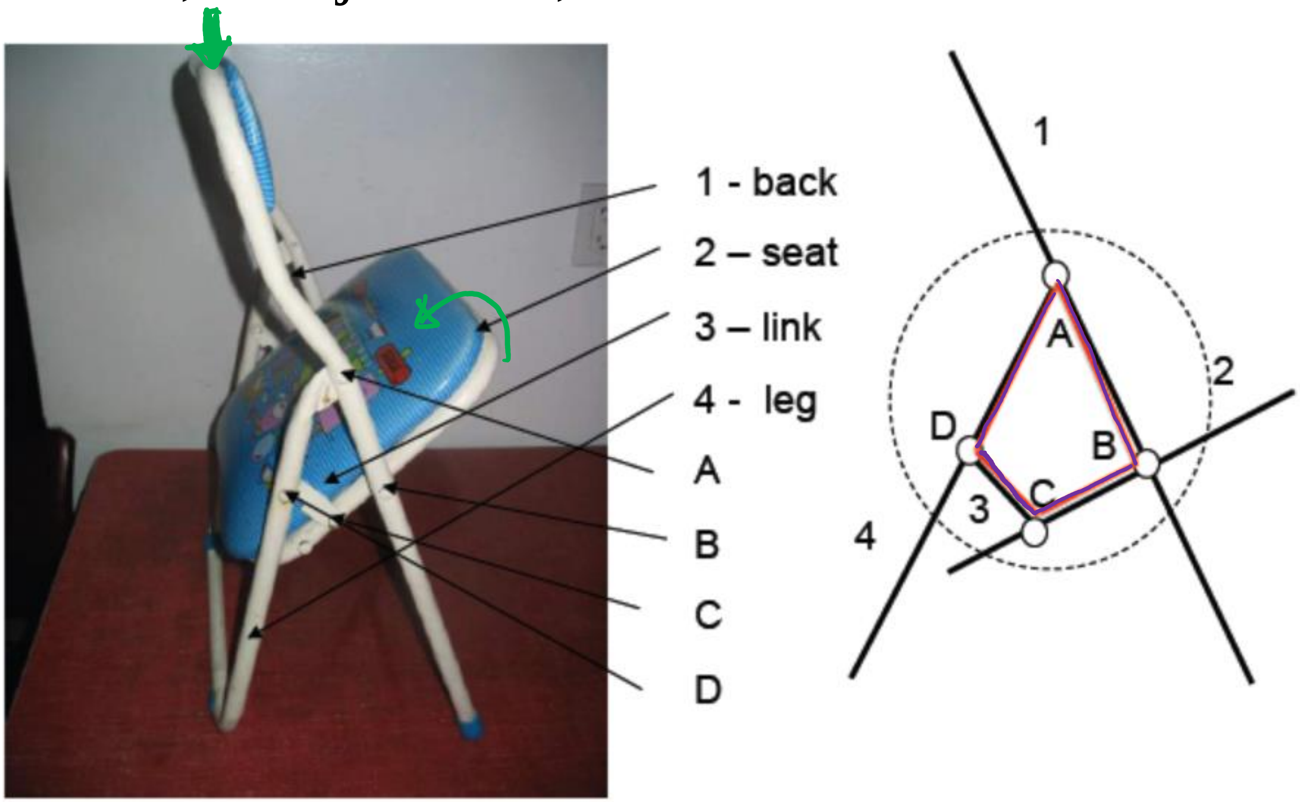
Steps for drawing kinematic diagrams

- Count the number of links and joints.
- Pick a link as the first link (number 1) and find all joints on it.
- Pick one joint (label A) on the first link that connects to one of its adjacent links.
- Do the same for the next link (number 2) with the next joint (label B).
- Continue till the last link (open chain), or back to the first link (closed chain).

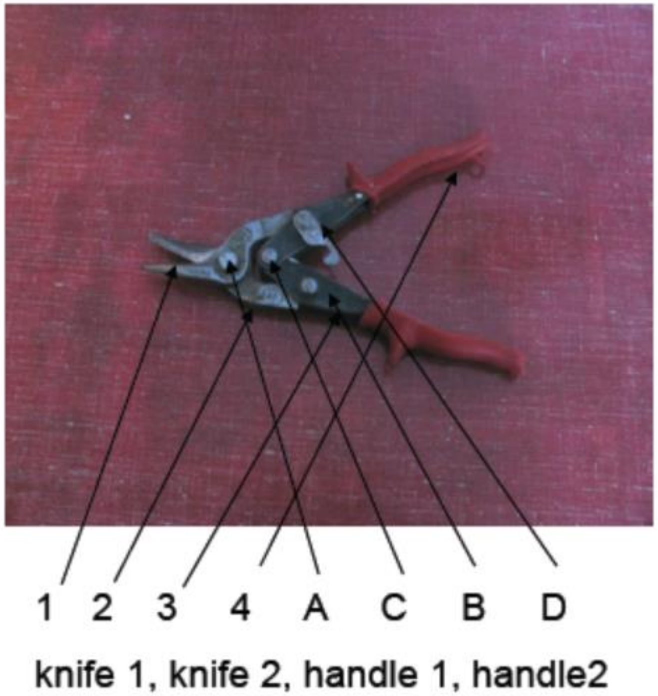
Folding chair example

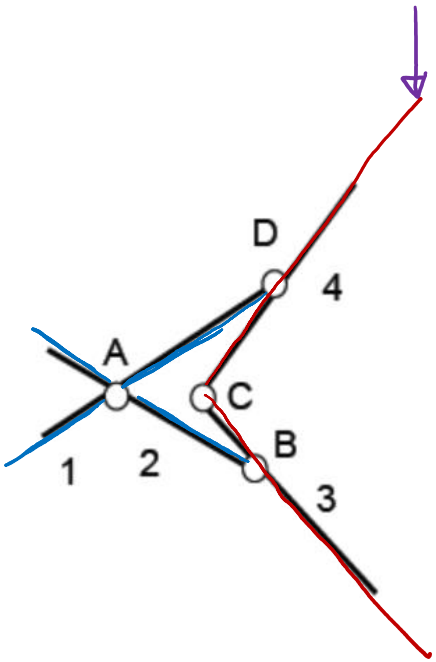
- # of links: 4; # of joints: 4; Red lines indicate the actual linkage



Scissor example

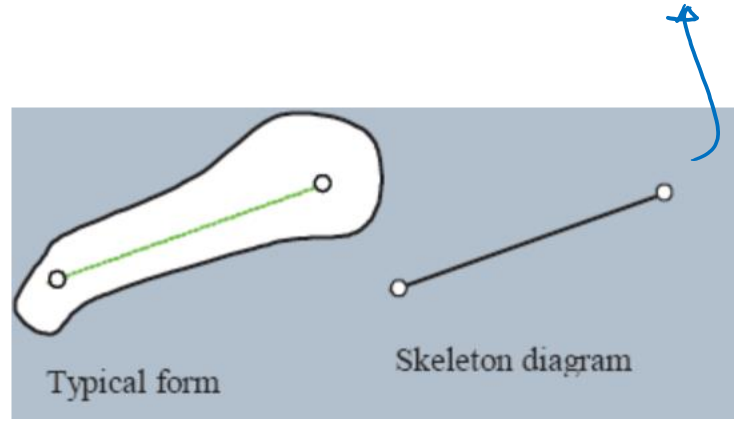
- # of links: 4; # of joints: 4.



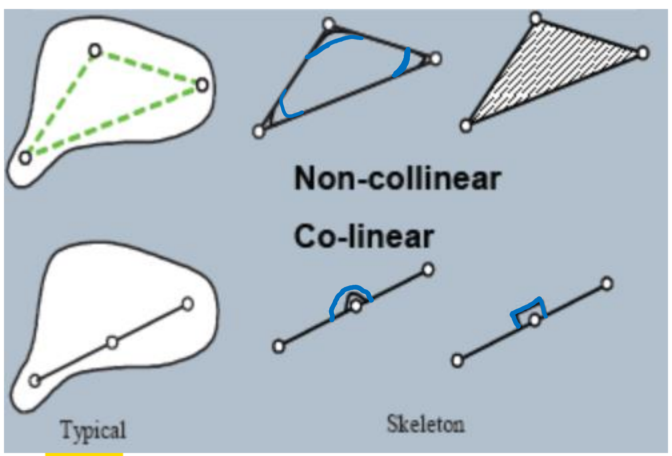


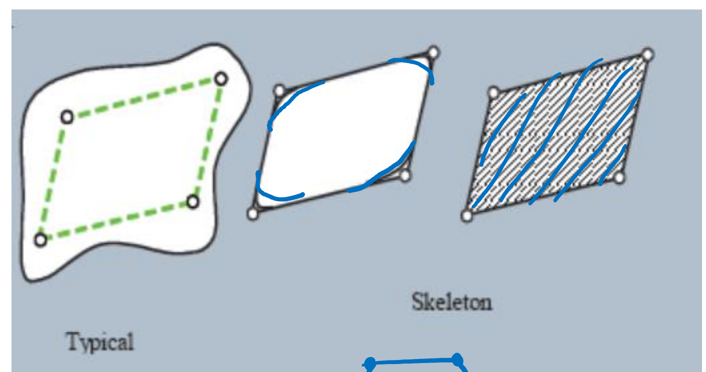
Planar link types

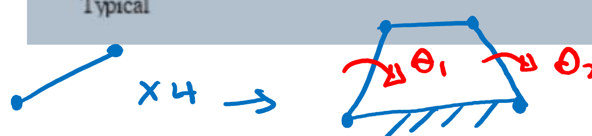
- A link has at least two joints.



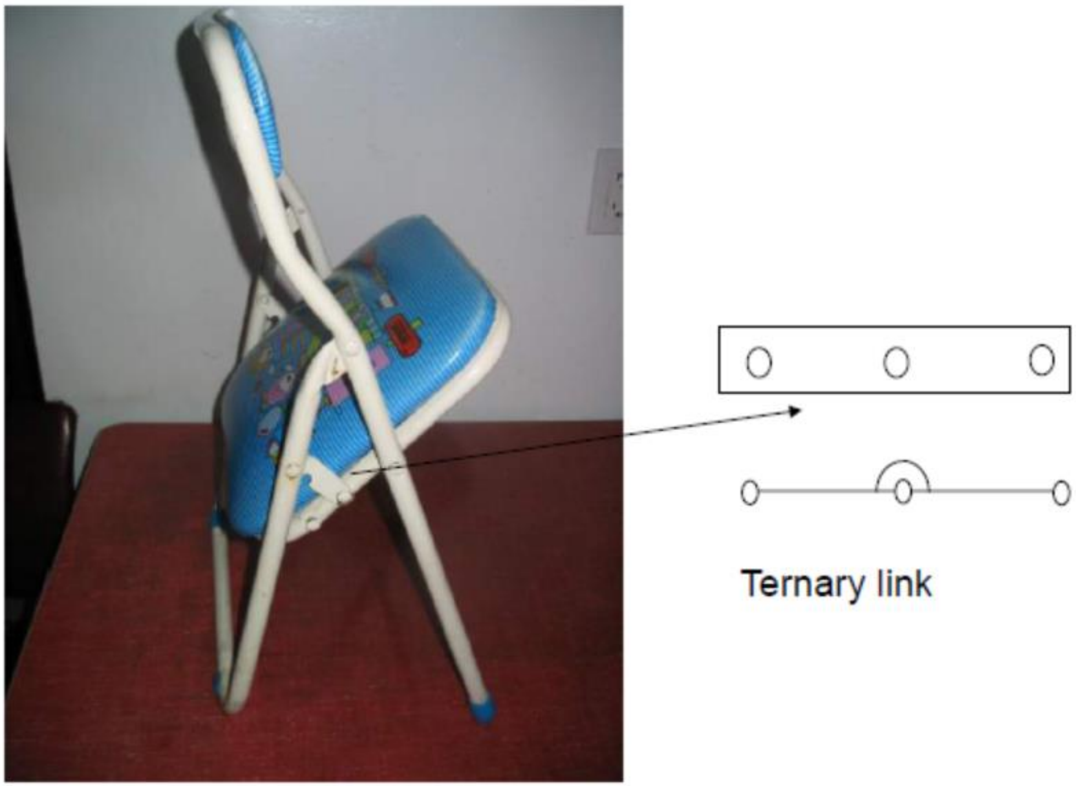
- Binary link: two joints.
- Ternary link: three joints.
- Quaternary link: Four joints.







Folding chair example

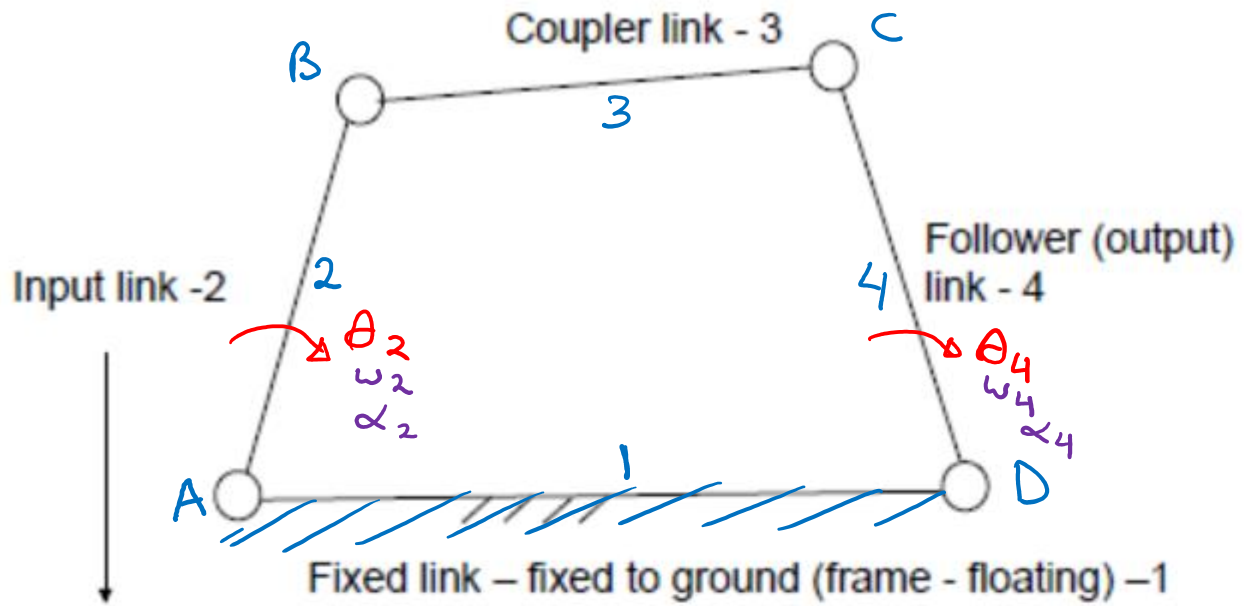


Linkages

- Linkages are classified in terms of the number of links.
- A dyad, two links with one joint, does not usually constitute a useful linkage (can be a part of a more complex mechanism).
- A triad, three links, does not usually form a useful linkage.
- A four-bar linkage is the most commonly used mechanism.

Linkages

- A four-bar linkage is the most commonly used mechanism.



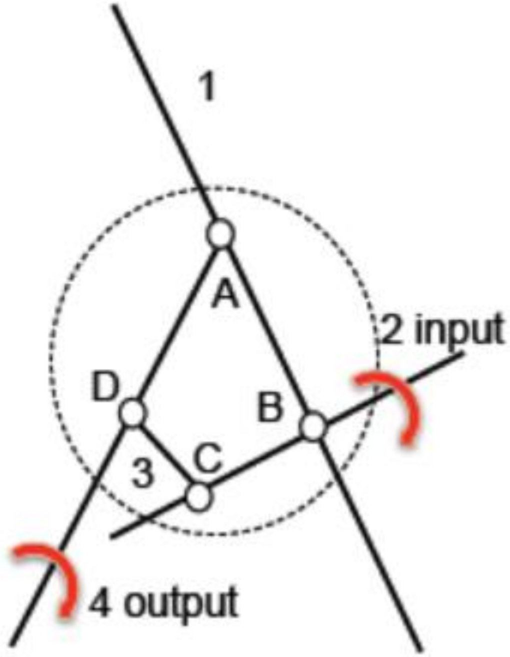
Position (motion) input: folding chair.

Force input: scissors.

Folding chair example

Link identification:

- Link 1 (back): frame (holding by one hand).
- Link 2 (seat): input link (moving by another hand).
- Link 3:couple link.
- Link 4 (back): follower link (following movement by hand).



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Linkage function classification

In terms of functions, four-bar linkages can be classified into:

- Function generator:

  The main concern is the relative motion or force between the input and the output link.

- Path generator:

  The main concern is the path of a tracer point of the coupler link (only the position is of interest).

- Motion generator:

  The main concern is the entire motion of the coupler link (both the position and orientation are of interest).

Folding chair example

Function classification:

1- Folding:

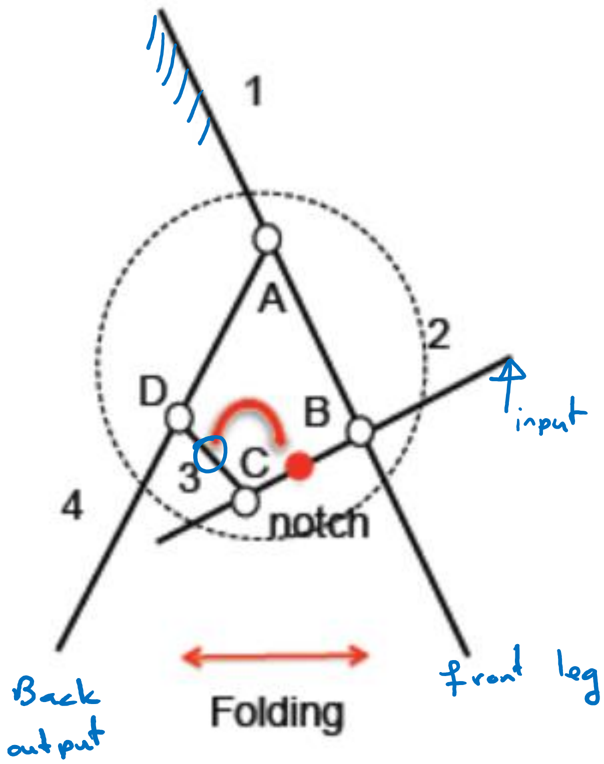
Function generation as it is concerned with the relative motion between 2 and 4.

2- Position the notch:

Path generation as it required the notch on the coupler (3) be placed on the locking position of the seat (2).

3-Locking:

Motion generation, as it requires the coupler (3) to be properly oriented for locking on the seat (2).



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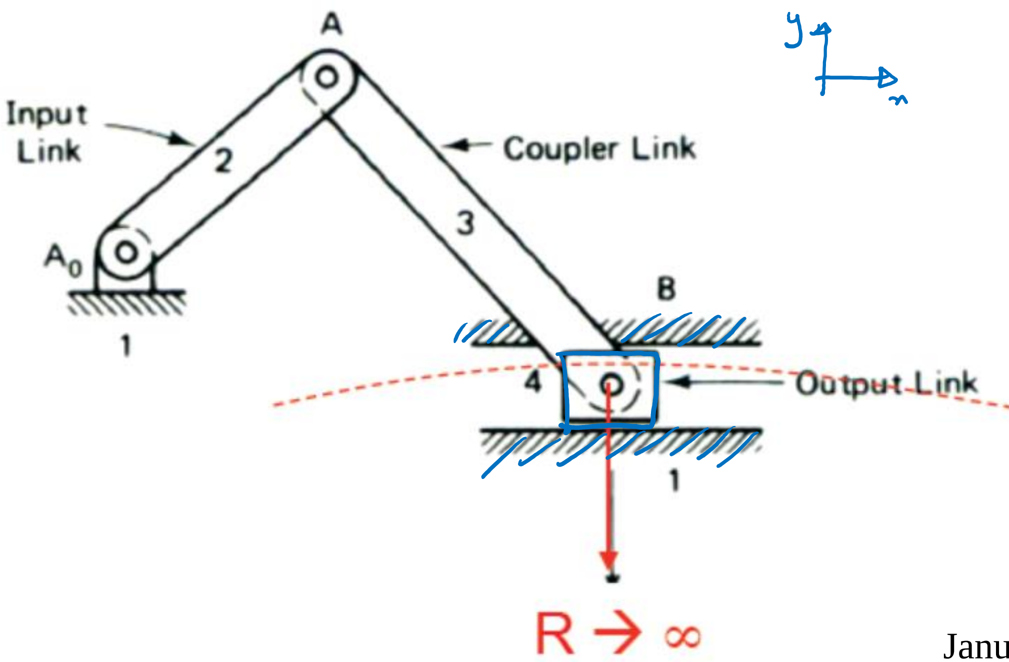
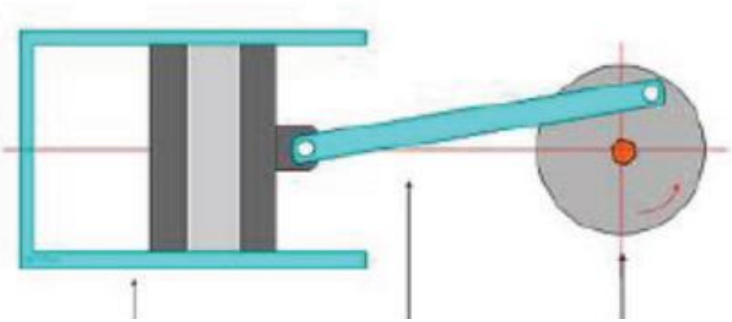
Crank-Slider / Slider-Crank

- The crank-slider is a special four-bar linkage in which the output link is replaced by a slider.
- The slider can be considered as an infinitely long output link.
- The slider is connected by a prismatic joint.



Crank-Slider examples

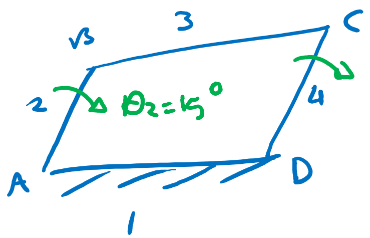
Home window



Conventional engine

Six-bar linkage (multiple loops)

- Four-bar linkage and slider-crank are single loops.
- A six-bar linkage is usually made of two four-bar linkages, also called two loop (chain) linkages.
- The same idea can be extended to form multiple loops.
- The six-bar linkage is used to extend the motion of a single four-bar linkage.



Six-bar linkage (multiple loops)

- Six-bar linkages can be classified by the way that two four-bar linkages are connected.
- The common six-bar linkages include:
  - Shared link.
  - Watt I and II.
  - Stephenson I, II, and III.

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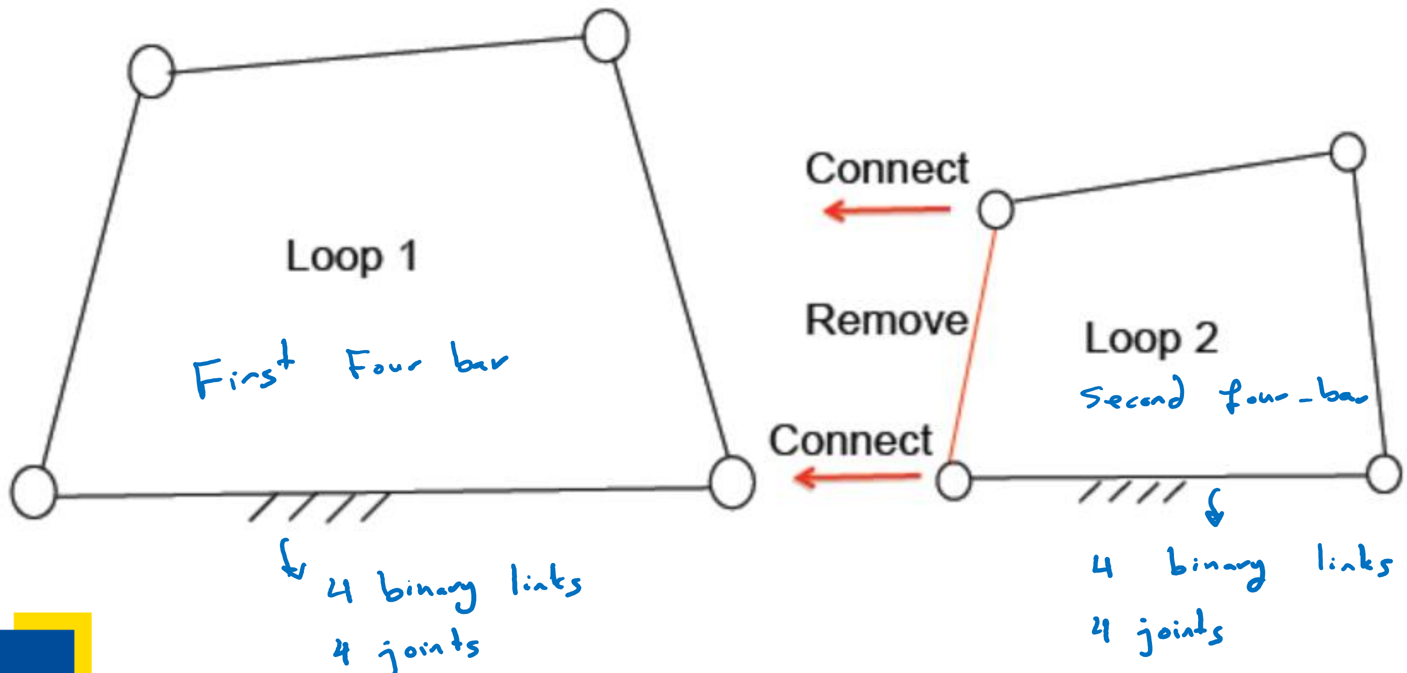
Six-bar linkage

- When loop 2 is connected to loop 1, the input link of loop 2 is removed and two of its joints are connected to loop 1.
- There are three ways for such a connection.

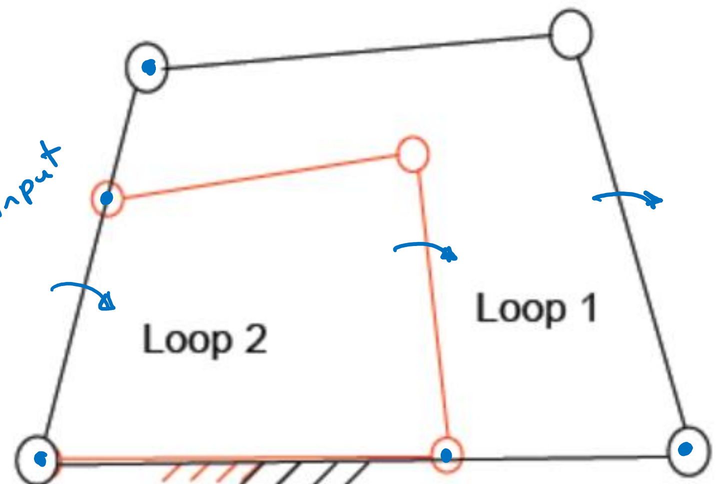


Shared link

- This is the case that the two joints of loop 2's input link are connected to the input link of loop 1.
- Both loops share the same input link and the frame.
- Both input link and frame are ternary links.

input of loop1 & 2 will move at the same time

input

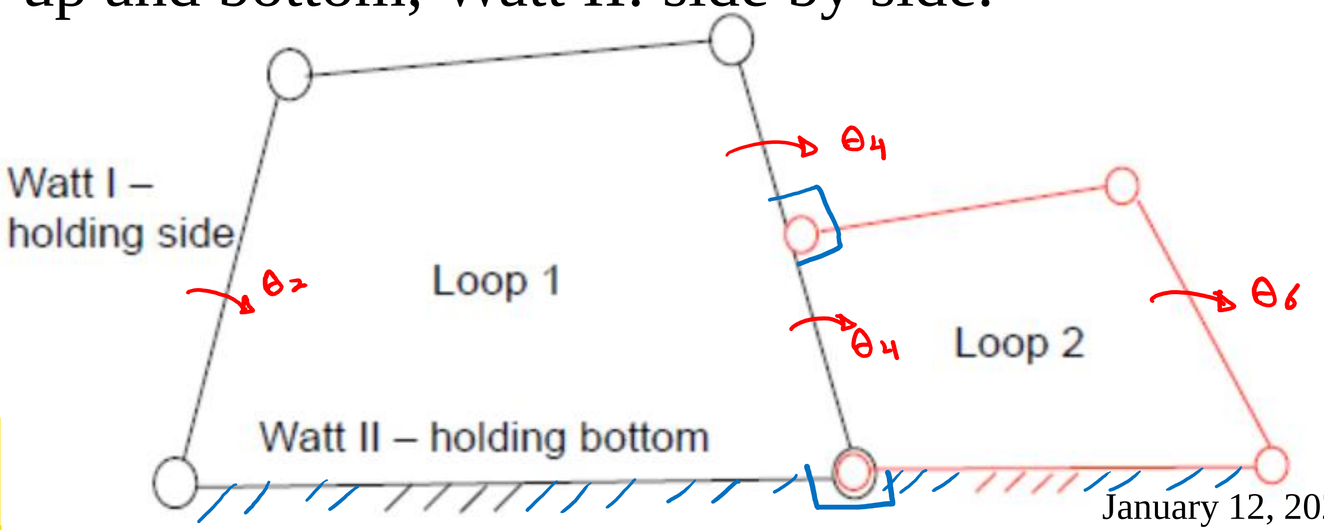


ground

6 links { 2 ternary links
4 binary links }

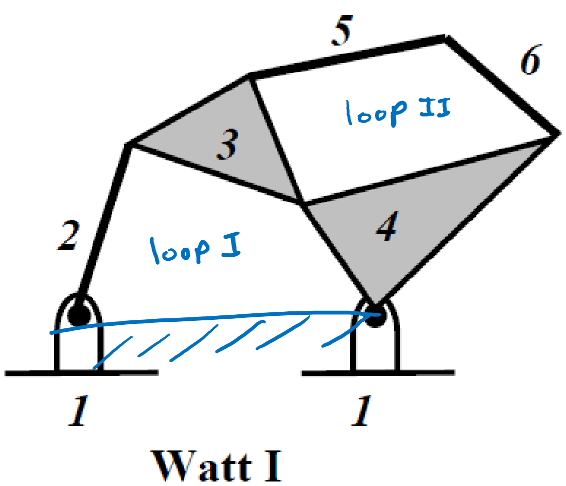
Watt type

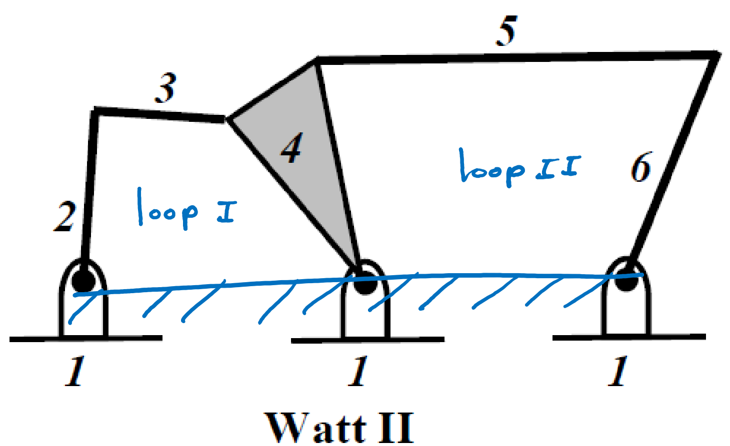
- This is the case that the two joints of loop 2's input link are connected to the output link of loop 1.
- The output link of loop 1 becomes the input link of loop 2, one ternary link.
- Both loops share the frame, another ternary link.
- Watt I: up and bottom; Watt II: side by side.



Watt Types

Watt II: ground is a ternary member

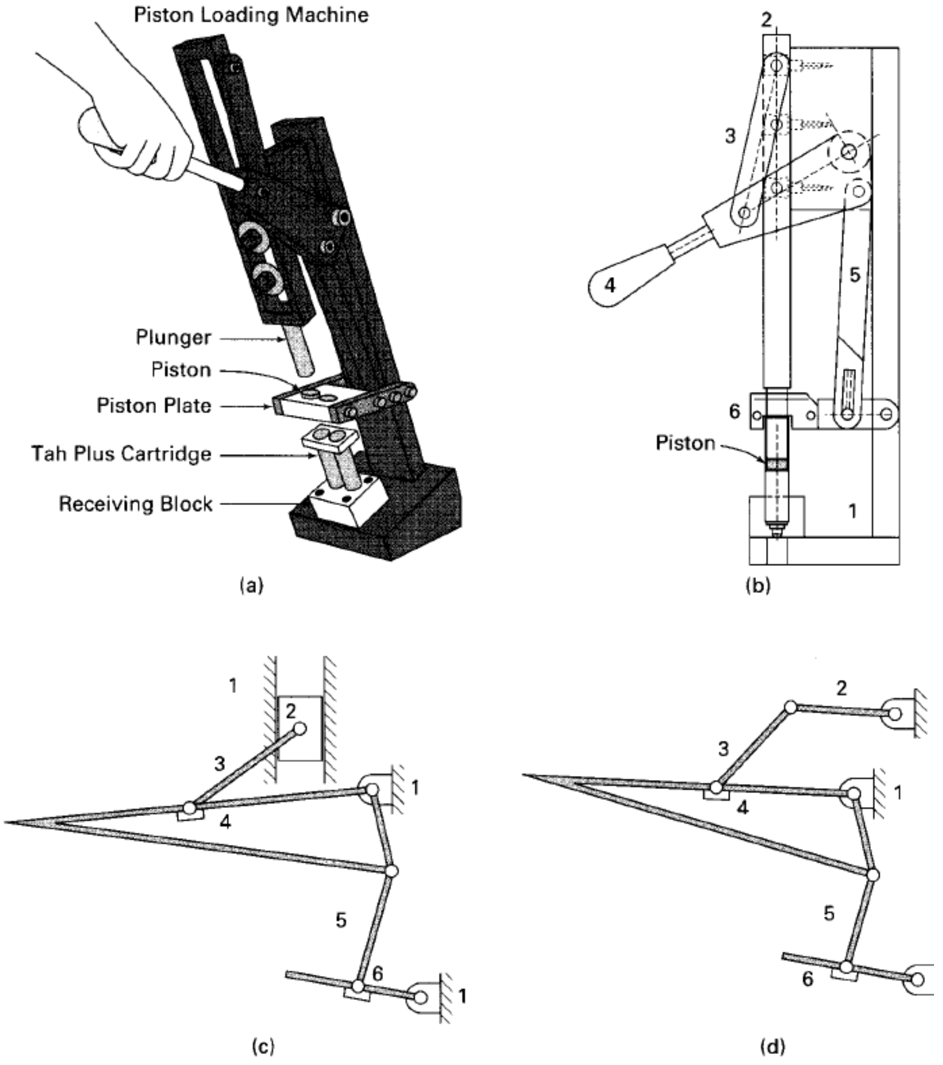




Example 1.4 from the textbook 1

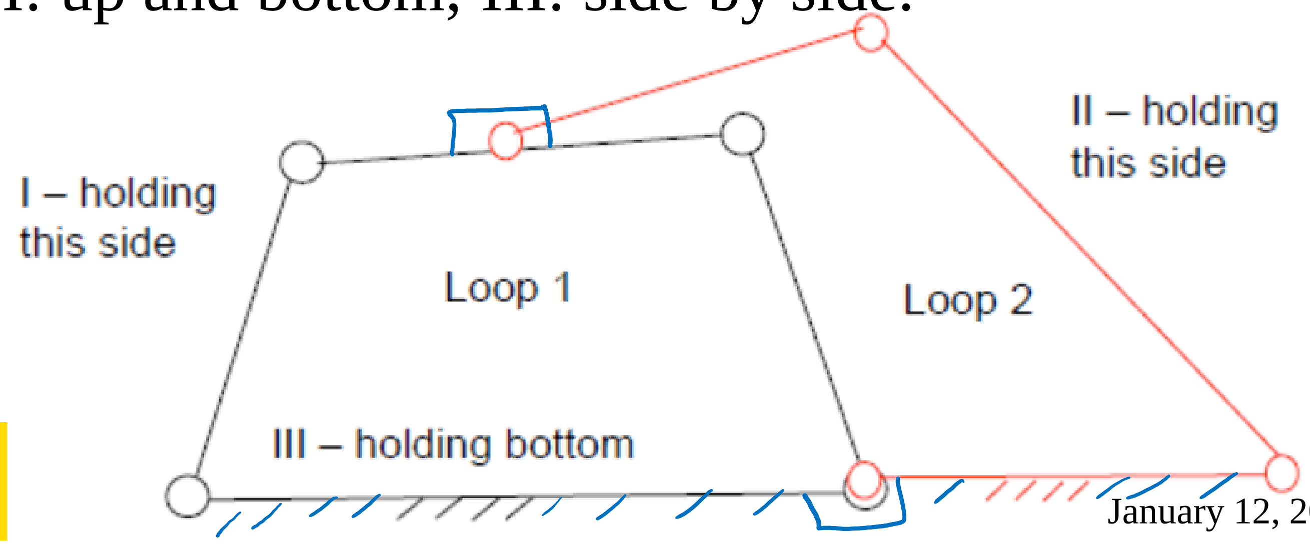
Hand-actuated mechanism:

Type Watt II



Stephenson type

- This is the case that the two joints of loop 2's input link are connected to the coupler of loop 1.
- The coupler of loop 1 becomes the input link of loop 2, one ternary link.
- Both loops share the frame, another ternary link.
- I and II: up and bottom; III: side by side.



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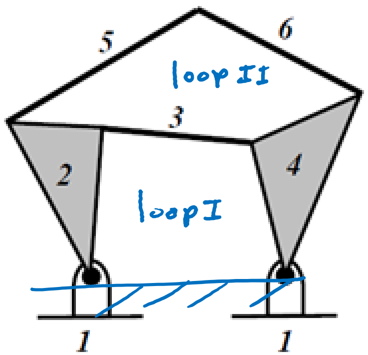
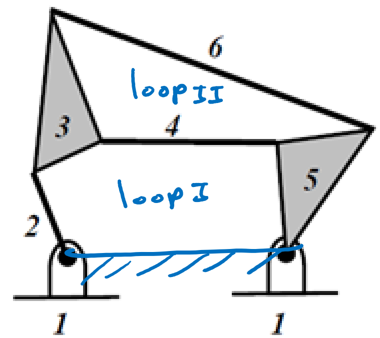
Stephenson types

- Stephenson I: a binary link is fixed connected to two ternary links. Stephenson II: a binary link is fixed and another one is connected to a ternary link.

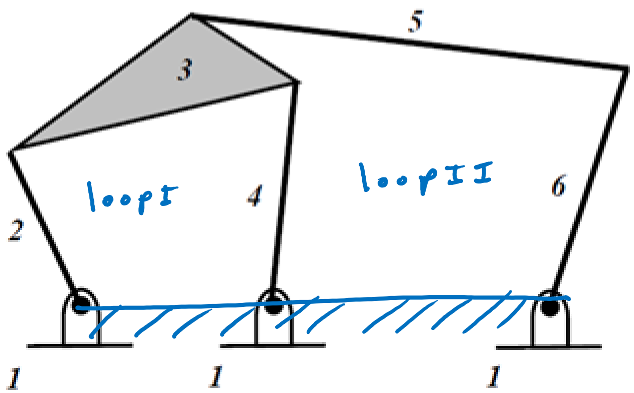


Stephenson I



Stephenson II

- Stephenson III: ground is a ternary link.



Stephenson III

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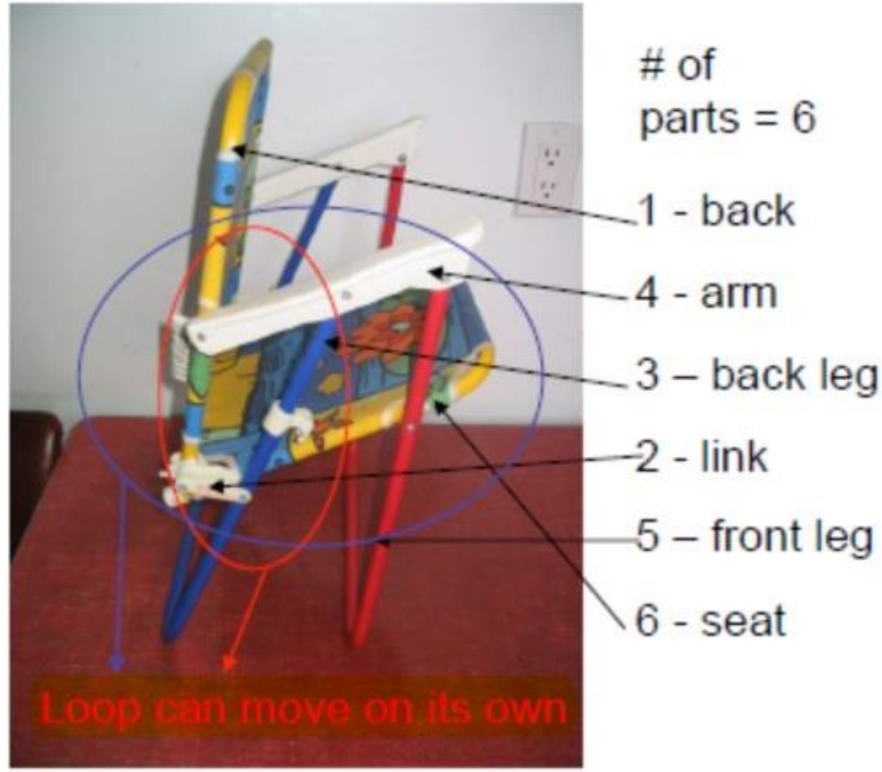
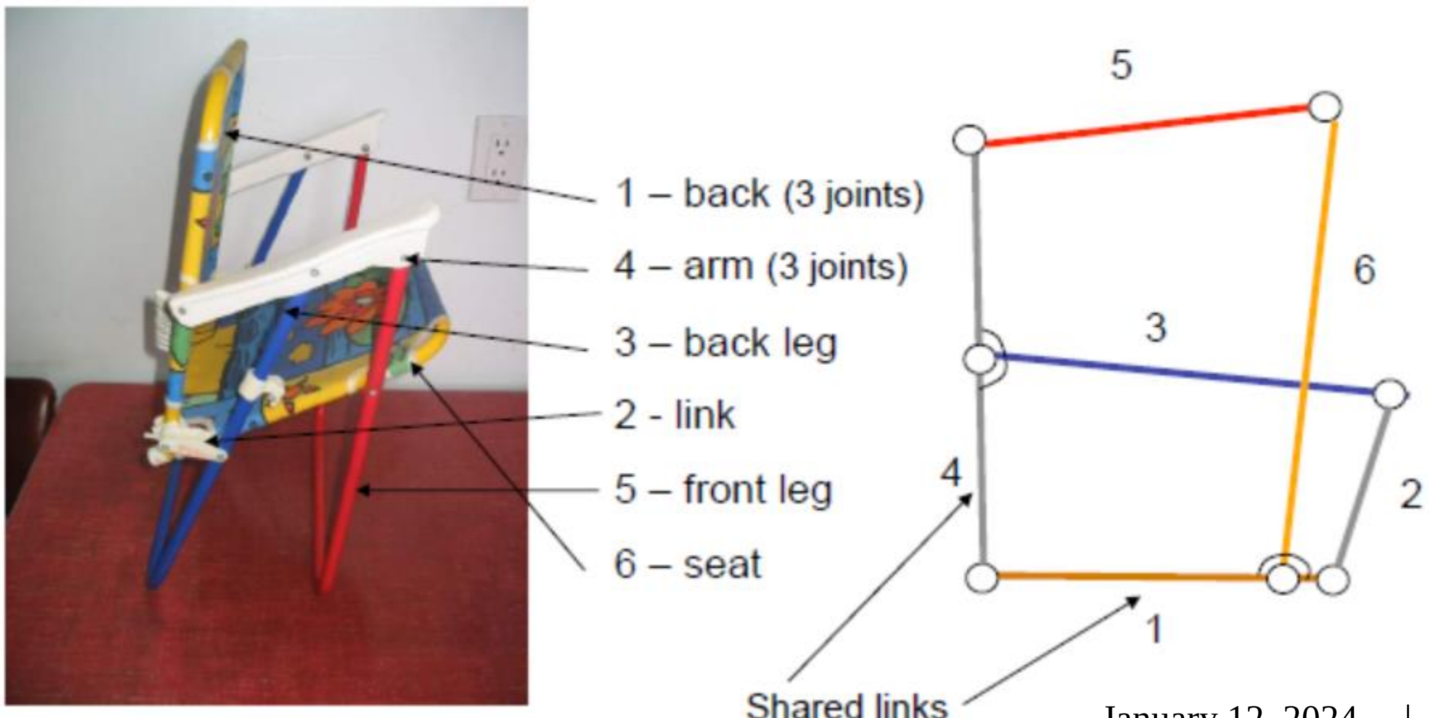
Folding chair-2 example



- What is the main difference compared to the four-bar chair?

Stick diagram for folding chair-2

- Steps: Identify a ternary link and try to form the first single loop and
- Links: 6; joints: 7; ternary links: 2.



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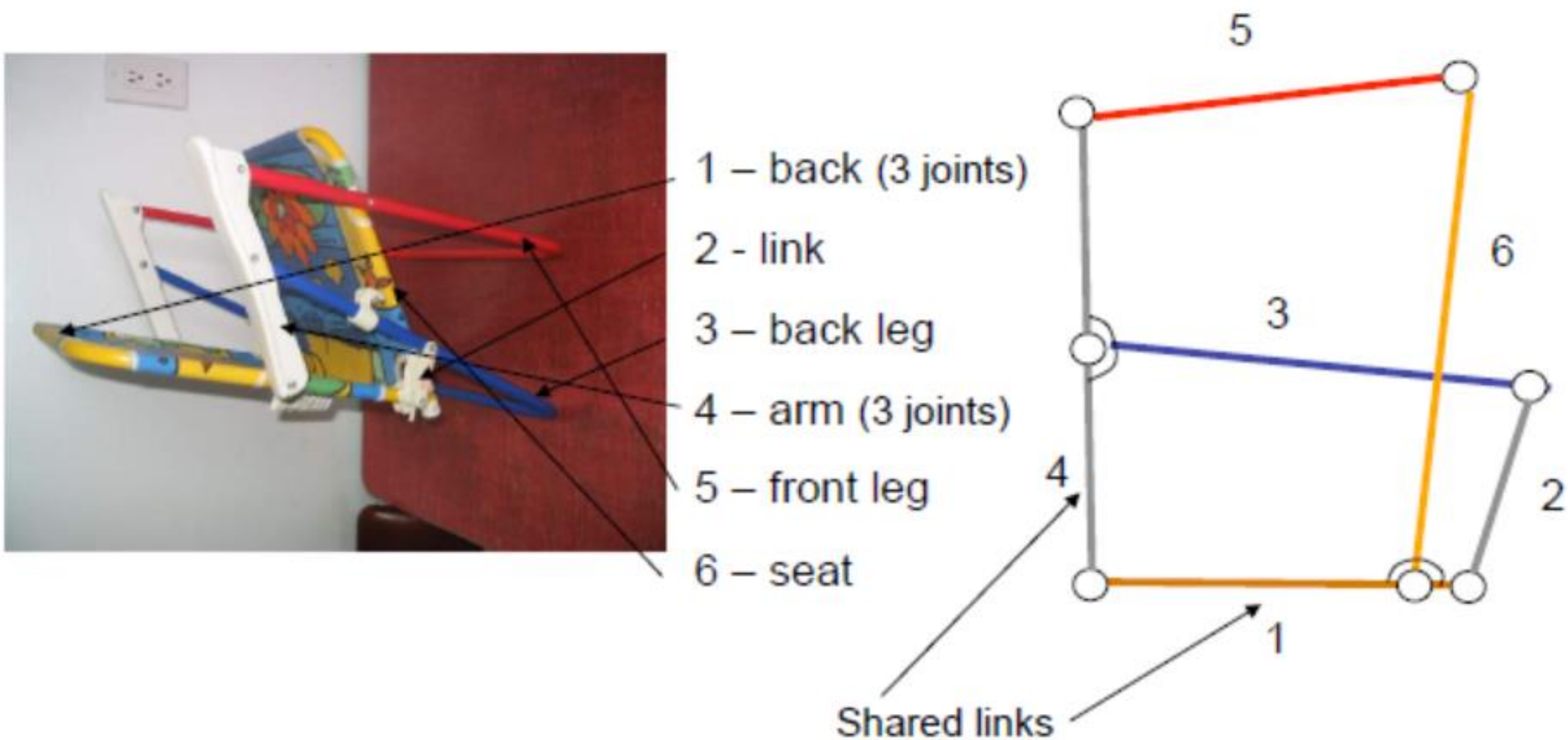
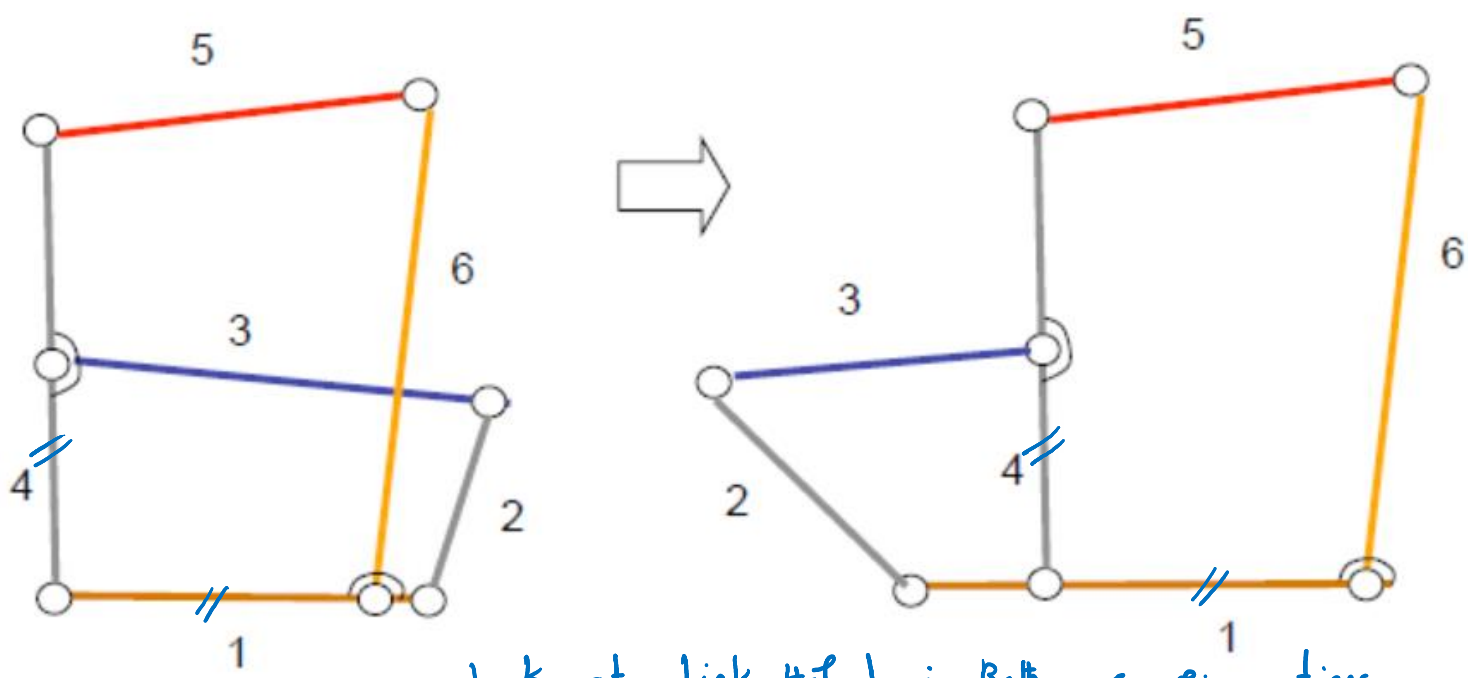
Stick diagram for folding chair-2



Stick diagram for folding chair-2

Hold back 1?

Hold seat 6?



look at link 4 & 1 in Both configurations

the same links are connected to them

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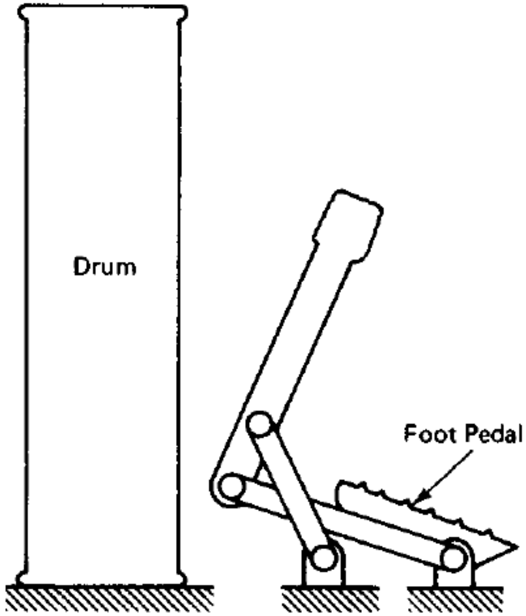
Assignment #1: Kinematic Diagram

Identify the types of the following mechanisms and draw kinematic diagrams.

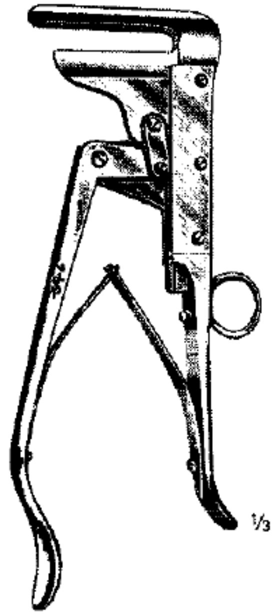
The mechanisms are selected from problem set 1 of the textbook #1

Solutions will be posted on the course website.

Problem 1.2 – drum foot pedal



Problem 1.3 – surgical tool



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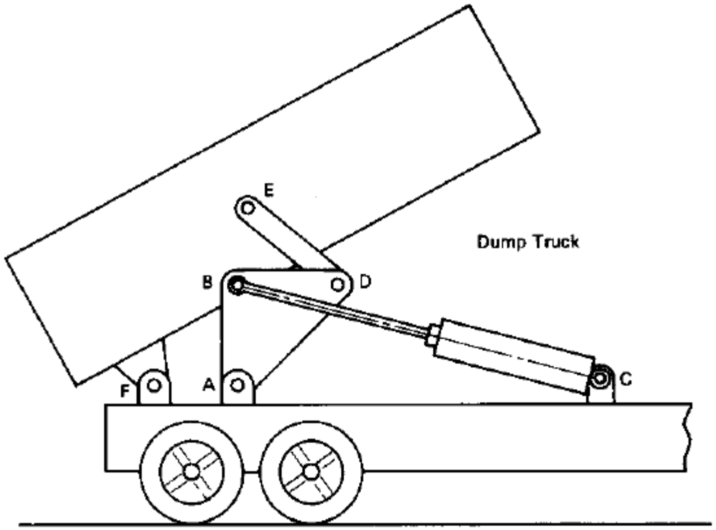
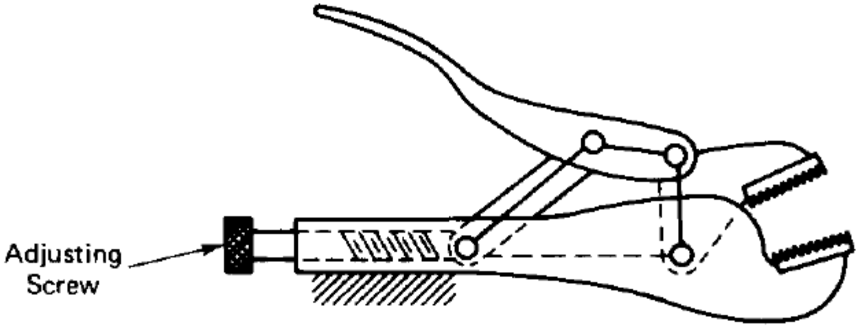
Assignment #1: Kinematic Diagram (Cont.)

Problem 1.4 – dump truck



Problem 1.6 - Pliers



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Assignment #1: Kinematic Diagram (Cont.)

Problem 1.19 – automobile hood